Write legibly showing all the steps VERY clearly. (Total points = 100).
Let $\Omega \neq \emptyset$ and $\mathcal{P}(\Omega)$ is the collection of all subsets of Ω, i.e the power set.

(1) (a) Define what it means for $\mathcal{C} \subset \mathcal{P}(\Omega)$ to be an algebra.
(b) Prove that $\mathcal{C}$ is an algebra *iff* $\mathcal{C}$ satisfies: $\Omega \in \mathcal{C}$ and $A, B \in \mathcal{C} \Rightarrow A \setminus B \in \mathcal{C}$
(c) Let $\Omega = \{a, b, c, d\}$ and $\mathcal{A}_1 = \{\phi, \Omega, \{a, b\}, \{c, d\}\}$, $\mathcal{A}_2 = \{\phi, \Omega, \{a, c\}, \{b, d\}\}$. Check $\mathcal{A}_1$ and $\mathcal{A}_2$ are algebras. Is $\mathcal{A}_1 \cup \mathcal{A}_2$ an algebra? Explain.
(d) Let $\mathcal{A}_n, n \geq 1$ be a sequence of *increasing* collection of algebras and $\mathcal{A} = \cup_{n=1}^{\infty} \mathcal{A}_n$. Prove that $\mathcal{A}$ is an algebra.

(5+15+15+15 =50 points)

(2) Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X : \Omega \to I\!R$ be a random variable.
(a) For $A, B \in \mathcal{F}$, if $P(A \Delta B) = 0$ then show $P(A) = P(B)$ $[A \Delta B = (A \setminus B) \cup (B \setminus A)]$.
(b) Let $B_i \in \mathcal{F}$ be such that $P(B_i) = 1$ for $i \geq 1$. Show $P(\cap B_i) = 1$.
(c) Argue $\{\omega : \sin(X(\omega)) + \cos(X(\omega)) = 1\}$ is in $\mathcal{F}$.

(3×10 = 30 points)

(3) Take $F(x) = x^3 \mathbf{I}_{(0,1)}(x) + \mathbf{I}_{[1,\infty)}(x), x \in I\!R$.
(a) Using the steps in Caratheodory Extension Theorem(s), show by starting from defining a measure μ_F on a suitable semi-algebra, how one can extend it to the Lebesgue-Stieltjes measure μ_F^* on $(I\!R, \mathcal{B}(I\!R))$.
[Write down the steps of the theorems carefully and state the results used in each step]
(b) Why is such a measure on $(I\!R, \mathcal{B}(I\!R))$ unique?
(c) If $A = (0.5, 2)$, then compute $\mu_F^*(A)$.

(10+5+5=20 points)